

INTRODUCTION

Project Title: AI Fit Track API

Team ID : SWTID-2026-7246

Team Size : 4

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1.Project Overview

- AI FitTrack API is an AI-based fitness management project developed to help users monitor, manage, and improve their daily fitness activities. The system collects important fitness information such as age, height, weight, activity level, exercise type, workout duration, daily steps, calories burned, and fitness goals.
- This information is processed through a REST API, which acts as a communication layer between the user interface, backend application, AI module, and database. The AI component analyzes the user's fitness data and provides personalized recommendations, such as suitable workout activities, exercise duration, and general fitness suggestions based on the user's goals and activity level.
- The project can be developed using Python with Flask or FastAPI, with SQLite, MySQL, or MongoDB as the database. A frontend can be developed using HTML, CSS, and JavaScript or React. Overall, AI FitTrack API provides a simple and structured platform for fitness tracking, data analysis, personalized recommendations, and progress monitoring through an API-based architecture.

2.Project Objectives

- To track fitness activities such as steps, workouts, exercise duration, and calories burned.
- To maintain user fitness profiles including basic details, activity levels, and fitness goals.
- To provide personalized recommendations using AI based on the user's fitness data and goals.
- To monitor fitness progress by maintaining workout history and activity records.

- To provide a REST API for communication between the frontend, backend, AI module, and database.
- To analyze fitness data and generate useful statistics to help users understand their activity patterns.
- To securely store fitness information in a database for future access and progress tracking.
- To encourage regular physical activity by providing simple, personalized fitness guidance.

3.Target Users

- Students – To track daily physical activities and workouts.
- Fitness Beginners – To get simple workout recommendations.
- Fitness Enthusiasts – To monitor exercise routines and progress.
- Regular Exercisers – To maintain workout records and activity history.
- Health-Conscious Individuals – To monitor their daily activity and fitness goals.
- Fitness Trainers – To manage and monitor users' fitness information.
- Developers – To integrate fitness tracking features into other applications

4.Pre-requisite & Knowledge requirements

- Students – To track daily physical activities and workouts.
- Fitness Beginners – To get simple workout recommendations.
- Fitness Enthusiasts – To monitor exercise routines and progress.
- Regular Exercisers – To maintain workout records and activity history.
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5.Project Architecture and Tech Stack

Project Architecture

- Frontend Layer – Allows users to enter fitness details, view workout information, and check recommendations.

- **API/Backend Layer** – Receives user requests, validates data, performs fitness calculations, communicates with the AI module, and sends responses.
- **Database Layer** – Stores user profiles, fitness activities, workout history, and progress data.

Architecture Flow:

- User → Frontend → REST API → Backend → AI Module → Database → API Response → User.

Technology Stack

Frontend : HTML,CSS,JavaScript

Backend : Python

Database : SQLite

API Testing : Postman

Code Editor : VS Code

6.Project setup and configuration

1. **Install Python** – Install Python on the computer and verify that it is working using `python --version`.
2. **Install VS Code** – Use Visual Studio Code to create and manage the project files.
3. **Create Project Folder** – Create a folder named `AI_FitTrack_API`.
4. **Create Virtual Environment** – Create a separate Python environment for the project to manage dependencies.
5. **Install Required Packages** – Install FastAPI, Uvicorn, SQLite-related libraries, and Scikit-learn if AI/ML functionality is included.
6. **Create Project Structure** – Organize the project into folders such as `api`, `models`, `database`, `services`, and `ai`.

7.Features and API Endpoints

Features

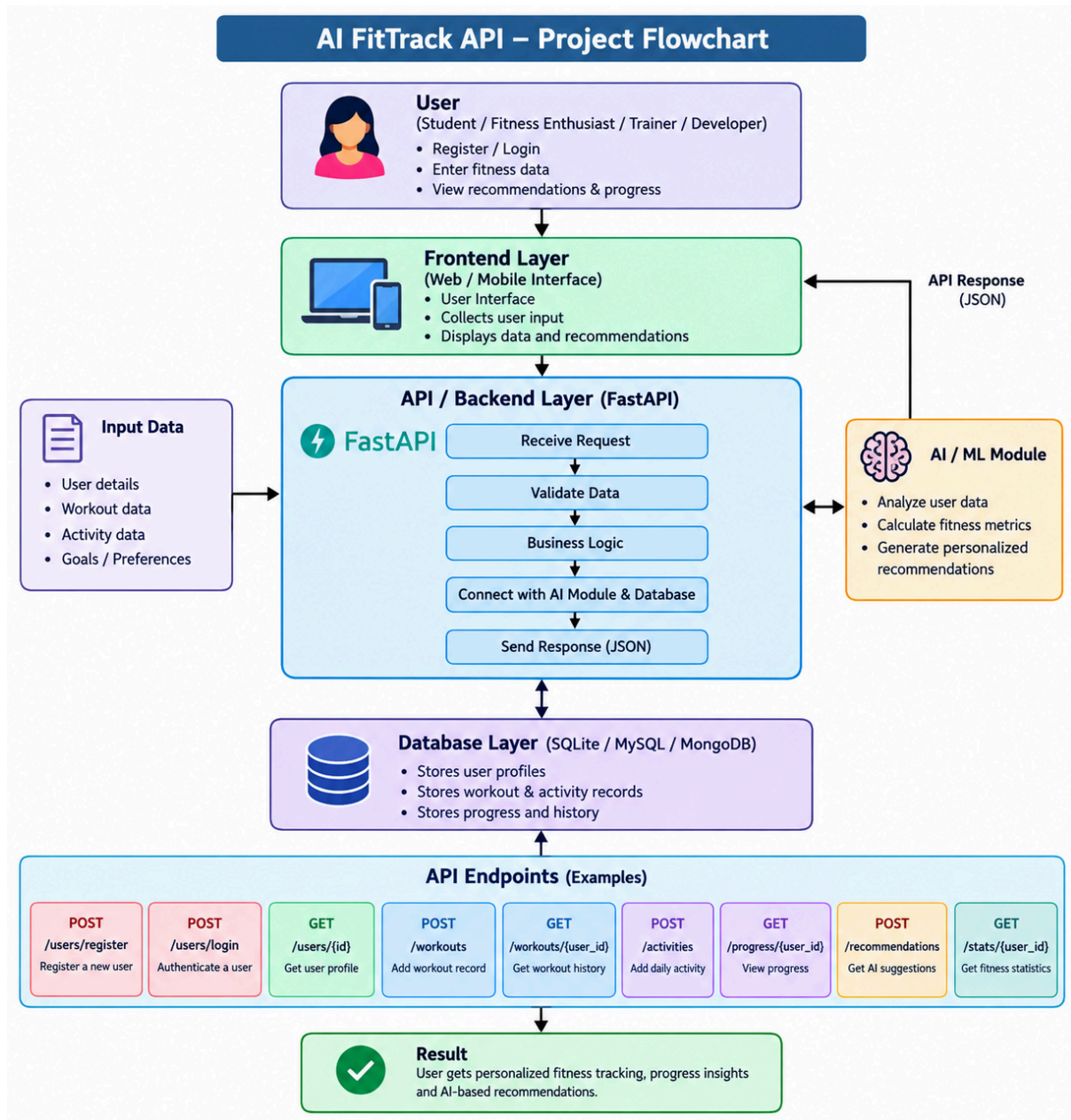
1. **User Registration & Login** – Allows users to create and manage their accounts.
2. **Fitness Profile Management** – Stores basic fitness details such as age, height, weight, activity level, and goals.

3. Workout Tracking – Records exercise type, duration, and workout date.
4. Daily Activity Tracking – Tracks steps, calories burned, and other activities.
5. AI Fitness Recommendations – Provides personalized workout suggestions based on user data.
6. Progress Tracking – Allows users to view their fitness activities and progress over time.
7. Fitness Statistics – Calculates useful information such as BMI, workout duration, and activity summaries.
8. Data Management – Stores and retrieves fitness records using the database.

API Endpoints

Method	EndPoints	Purpose
POST	/users/register	Register a new user
POST	/users/login	Authenticate a user
GET	/users/{id}	Get user profile
PUT	/users/{id}	Update user profile
POST	/workouts	Add a workout record
GET	/workouts/{users_id}	Get user's workout history
PUT	/workouts/{id}	Update a workout
DELETE	/workouts/{id}	Delete a workout
POST	/activities	Add daily activity
GET	/activities/{users_id}	View activity records
GET	/progress/{user_id}	View fitness progress
POST	/recommendations	Generate AI fitness recommendation

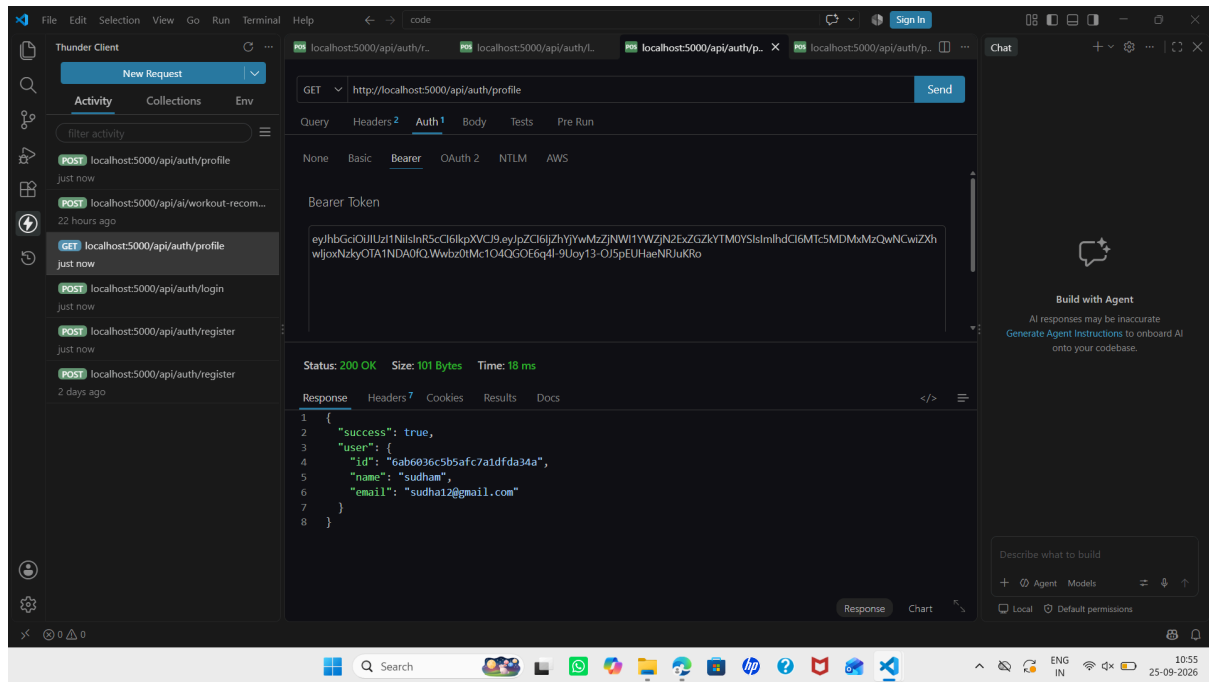
8. Project Floechart



9. Output

The screenshot shows the Thunder Client interface with a POST request to `http://localhost:5000/api/auth/login`. The request body is a JSON object: `{ "email": "sudha12@gmail.com", "password": "sudha3456" }`. The response status is `200 OK` with a size of `283 Bytes` and a time of `108 ms`. The response body is a JSON object: `{ "success": true, "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZCI6IjZyYjVhZzJmMTI1YmZjN2ExZGZkYTM0YSIsIm1ldCI6MTc5MDMwMzQwIiwiaWF0IjoxNzkyOTA1NDABFQ.wwz8tMc104Q50Eeq41-9Uoy13-0T5pELHaenR3JukRo", "user": { "id": "6ab6036c5b5afc7a1dfa34a", "name": "sudham", "email": "sudha12@gmail.com" } }`. The interface includes a sidebar with a list of requests, a main panel for request details, and a chat panel on the right.

The screenshot shows the Thunder Client interface with a POST request to `http://localhost:5000/api/auth/register`. The request body is a JSON object: `{ "name": "sudham", "email": "sudha12@gmail.com", "password": "sudha3456" }`. The response status is `201 Created` with a size of `283 Bytes` and a time of `193 ms`. The response body is a JSON object: `{ "success": true, "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZCI6IjZyYjVhZzJmMTI1YmZjN2ExZGZkYTM0YSIsIm1ldCI6MTc5MDMwMzQwIiwiaWF0IjoxNzkyOTA1MzI0FQ.jx6oTkfyK2SHDvIemybXl381UaT5wThFRx5CwB00FM", "user": { "id": "6ab6036c5b5afc7a1dfa34a", "name": "sudham", "email": "sudha12@gmail.com" } }`. The interface includes a sidebar with a file explorer, a main panel for request details, and a chat panel on the right.



10.Conclusion

The AI FitTrack API provides a simple and efficient platform for managing and monitoring fitness activities. The system allows users to maintain their fitness profiles, record workouts and daily activities, track their progress, and receive personalized recommendations using AI. By combining Python, FastAPI, AI/ML, and a database, the project provides a structured and scalable solution for fitness management. The API-based architecture also makes it easier to integrate the system with web or mobile applications. Overall, AI FitTrack API helps users understand their fitness activities and make better, data-based decisions about their workout routines.